

RAGtune Experiment Grid Report

Detailed systematic ablation study across quality metrics (NDCG@5, Recall@5, MRR) and latency.

Methodology & Configuration

1. Budget Levels

Definitions for document reranking constraints and batch sizes:

Level	Rerank Docs	Batch Size	Description
Tight	5	2	Minimum cost, low latency constraint.
Medium	15	5	Balanced configuration for general use.
Loose	30	10	Aggressive reranking for maximum precision.

Dataset: SciFact

~5K docs, 50 queries, binary qrels (BEIR)

Group	Config	NDCG@5	Recall@5	MRR	Latency (ms)	Rerank Docs
A	bm25_only	0.7931	0.854	0.773	8	0
B	monot5_tight	0.8536	0.854	0.8547	605	5
B	monot5_medium	0.8458	0.858	0.8485	1678	15
B	monot5_loose	0.8471	0.859	0.8484	3324	30
C	baseline_est	0.8458	0.858	0.8485	1695	15
C	similarity_est	0.8458	0.858	0.8485	1689	15
C	reformir_est	0.8458	0.858	0.8485	1706	15



Dataset: TREC-COVID

50K docs, 20 queries, graded qrels (ir-datasets)

Group	Config	NDCG@5	Recall@5	MRR	Latency (ms)	Rerank Docs
A	bm25_only	0.5908	0.004	0.5588	36	0
B	monot5_tight	0.663	0.004	0.6421	502	5
B	monot5_medium	0.7448	0.0051	0.7583	1486	15
B	monot5_loose	0.699	0.0046	0.7081	3067	30
C	baseline_est	0.7448	0.0051	0.7583	1497	15
C	similarity_est	0.7448	0.0051	0.7583	1497	15
C	reformir_est	0.7448	0.0051	0.7583	1502	15